

Miguel G. SILVA

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SUMMARY

PhD researcher and applied AI scientist. I work where research and engineering meet: framing the hard modeling problem (e.g., neural surrogates for physics simulation, multi-objective constrained optimization over engineered products), then carrying the answer into production. Strong background in time-series modeling, pattern discovery, and interpretable ML, with first-author publications in Q1 journals.

WORK EXPERIENCE

2025 – PRESENT	Applied AI Scientist at DAREDATA ENGINEERING - Co-led the design of a production agentic engineering-to-order loop for a global automotive cable manufacturer, replacing a 16-week / ~€10k physical prototype-and-lab cycle with a 1-day / ~€1.8k software workflow (~80× faster, 5–10× cheaper per RFP). - Formulated and built the surrogate-modeling layer: a Graph Neural Network with calibrated uncertainty, trained on multiphysics simulation history and turning hours of physics solves into millisecond inference. - Designed the multi-objective constrained optimization over a mixed continuous/discrete design space (material × geometry × stack), balancing cost and performance under regulatory and manufacturability constraints. Calls the surrogate as a fast oracle, routing survivors to FEM validation. - Partnered with the client on product direction: framed engineering requirements as research problems, defined the tools and skills exposed by the engineer assistant, and informed roadmap trade-offs across the loop.
2021 – 2026	Doctoral Researcher at LASIGE & INESC-ID - Analyzed smart meter data from 2,170 households using biclustering, uncovering statistically significant and actionable local water consumption patterns inaccessible to traditional clustering methods. - Developed a scalable methodology to extract population density patterns from mobile phone data (3,743 cells) across Lisbon, uncovering multi-level seasonality and emerging urban trends. - Designed a methodology to extract recurring, non-trivial, and statistically significant patterns in multivariate time series data. - Published 3 first-author papers in top-quartile (Q1) scientific journals.
2022 – 2024	Teaching Assistant at IST, UNIVERSITY OF LISBON - Selected as invited TA to deliver 140+ hours of lectures and labs on Information Retrieval (Python) and Database Systems (SQL) to over 300 students. - Maintained an average student rating of 4.5/5 across three semesters.

EDUCATION

2021 - EXPECTED 2026	PhD in Computer Science <i>Faculty of Sciences University of Lisbon, Portugal</i>
2018 - 2020	MSc in Data Science <i>Faculty of Sciences, University of Lisbon, Portugal</i> GPA: 18 / 20 , THESIS: 19/20

PROJECTS

2026	Statistical Significance for Multivariate Time Series Motifs <i>Pattern Recognition Letters</i> PyPI GitHub Open-sourced <code>msig</code> , a Python package implementing statistical significance criteria that separate genuine recurring structure from spurious patterns in multivariate time series motif analysis.
2025	Motif Mining in Multivariate Time Series <i>Publication</i> GitHub Built an unsupervised pipeline to mine recurring patterns in residual time series, revealing actionable behavioral insights in energy use, human mobility, and urban activity.
2025	Next-Motif: Predicting Irregular Pattern Recurrence in Time Series <i>Publication</i> GitHub Leveraged LSTM, TCN, and Transformer models to build a forecasting system for irregular event patterns in time series, reducing prediction error by 21–24% on urban mobility and electricity consumption data.
2024	UrbanFlow: Spatiotemporal Analysis of Population Density Data <i>Publication</i> GitHub Developed a Python-based, Docker-deployable tool to extract and visualize spatiotemporal population trends from mobile network data (3,743 cells), enabling actionable insights for data-driven urban planning.
2023	Biclustering Water Consumption Patterns from Smart Meter Data <i>Publication</i> Applied biclustering to household water data from a luxury resort (2,170 users), uncovering high-resolution seasonal consumption profiles to support anomaly detection and data-driven resource planning.

TECHNICAL SKILLS

Languages & Core: Python · SQL · Git · LaTeX

ML / Deep Learning: PyTorch · Scikit-learn · Optuna · Pandas · NumPy

Applied AI: Surrogate Modeling · Multi-Objective Optimization · Time-Series Modeling · Uncertainty Quantification · Agentic AI

Engineering & MLOps: FastAPI · Pydantic · Docker · CI/CD (GitHub Actions) · MLflow · Async Pipelines · ETL